# Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system under nonnegative adversarial arrivals, convex state–action costs, and state-dependent action constraints. At each time, the controller chooses a feasible action before the current arrival and cost are revealed. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained finite-memory policy class whose fixed policies satisfy the per-step action constraint by construction. For online adaptation, an entropic online mirror descent (OMD) update maintains the SDAC parameters on the subprobability simplex, while scalar action clipping ensures feasibility of the actual time-varying trajectory. We prove $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy, where $H$ is the memory horizon. We further show that SDAC geometrically approximates every feasible proportional state-feedback policy; choosing $H=\Theta(\log T)$ yields $O\left(\sqrt{T\log\log T}\right)$ regret against the best policy in this class. The setting is motivated by energy-harvesting battery management and related storage systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

## I. Introduction

### A. Motivation and Problem Setting

MANY dynamic resource systems can be modeled by a storage state that evolves over time and constrains the control action at every time. The controller must choose an action subject to a state-dependent action constraint before observing the corresponding arrival and cost.

This creates a constrained discrete-time control problem with state–action coupling: aggressive early actions can harm later performance, while overly conservative actions can increase the stage cost. Our objective is to remain feasible at every time while competing with the best fixed policy in a tractable class that approximates feasible proportional state-feedback policies.

### B. Model and Information Structure

We study a discrete-time storage system with state $x_t \geq 0$, where $x_t$ models the storage level at the beginning of time $t$. At each time $t$, the controller chooses a control action $u_t$ under the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t,$$

where $\alpha \in (0,1)$ is a fixed known retention coefficient (so a fraction $1-\alpha$ of the state is lost each period). After the action is chosen, a nonnegative adversarial arrival $w_t$ (a resource inflow) is revealed, and the state evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t.$$

Although this input is commonly called a disturbance in the online-control literature, we call $w_t \in [0,1]$ an *arrival* to emphasize its nonnegativity; we retain the standard names Disturbance-Action Control (DAC) and Simplex Disturbance-Action Control (SDAC). Nonnegativity is essential to the storage interpretation and to the feasibility properties of SDAC.

Unlike stochastic models, our formulation is adversarial and non-anticipatory. At time $t$, the environment may select $w_t$ and $c_t$ using the history through time $t-1$, but not the current action; the controller then chooses $u_t$, after which $w_t$ and the entire function $c_t$ are revealed. The goal is low cumulative cost relative to the best fixed policy in a class defined below, with both the online and benchmark trajectories remaining feasible.

### C. Application Domains

This abstraction captures a broad set of domains:

- **Energy harvesting and battery management:** $x_t$ is battery state of charge (retention coefficient $\alpha$ after self-discharge), $w_t$ is harvested energy, and $u_t$ is discharge or consumption bounded by available energy $\alpha x_t$.
- **Queueing networks and buffer management:** $x_t$ is queue/buffer occupancy, $w_t$ is packet arrivals, and $u_t$ is service bounded by $\alpha x_t$.
- **Inventory control:** $x_t$ is on-hand inventory (retention coefficient $\alpha$ after spoilage/deterioration), $w_t$ is supply or replenishment, and $u_t$ is sales bounded by $\alpha x_t$.
- **Water-reservoir management:** $x_t$ is stored water (retention coefficient $\alpha$ after evaporation/seepage), $w_t$ is inflow, and $u_t$ is controlled release bounded by $\alpha x_t$.

We take energy harvesting as the running interpretation; the other domains share the same storage dynamics and state-dependent action constraint.

### D. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a DAC-inspired policy class for storage dynamics with the state-dependent action constraint $0 \leq u_t \leq \alpha x_t$. We prove that every fixed SDAC policy is feasible, its induced loss is convex and Lipschitz in the policy parameter, and the class geometrically approximates feasible proportional state-feedback policies.

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- **Online control and regret.** We develop an entropic OMD controller over SDAC parameters. A Bregman projection maintains the simplex constraint on the parameters, and scalar action clipping maintains feasibility of the actual trajectory. The controller achieves $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy and, with $H = \Theta(\log T)$, achieves $O\left(\sqrt{T\log\log T}\right)$ regret against the best feasible proportional state-feedback policy.

### E. Related Work

*a) Online nonstochastic control:* A recent line of work studies online control of linear dynamical systems under adversarial disturbances and costs. Its performance criterion is cumulative regret relative to a benchmark policy class chosen in hindsight, under suitable stability conditions [1]–[3]; see [4] for an overview.

A central idea in this literature is *Disturbance-Action Control* (DAC),[1] a policy parameterization that combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard assumptions, this representation yields tractable surrogate losses in the policy parameters and supports efficient no-regret updates. Extensions include partial observability [5] and bandit feedback [6].

Constrained online-control methods have also been developed for affine and general static state–input constraints [7], [8]. Our focus is the storage-specific constraint $0 \le u_t \le \alpha x_t$, whose feasible action set depends on the current state. Exploiting nonnegative arrivals, SDAC uses a simplex parameterization to make every fixed policy feasible; scalar action clipping preserves feasibility when the parameter changes online.

*b) Energy harvesting:* Lin et al. [9] study online age-of-information optimization in an energy-harvesting system with related battery dynamics. They compare with an unrestricted offline optimum through competitive ratio; their objective depends on the energy allocation but not directly on the battery state, and the current input is revealed before allocation. In our model, the controller acts before observing the current arrival and cost, the cost may depend jointly on state and action, and performance is measured by regret against a fixed policy class.

Yu and Neely [10] study expected utility maximization under i.i.d. system states. Asgari and Neely [11] express the guarantee as an $O(\sqrt{T})$ regret–battery-capacity tradeoff and improve its dependence on the decision dimension, allowing arbitrary loss sequences while retaining i.i.d. energy arrivals. Both works use action-only objectives and fixed-decision comparators. Here, arrivals are adversarial, costs depend on both state and action, and the comparator is a fixed state-responsive policy.

*c) Queueing and network optimization:* Queueing control provides a mathematically related perspective based on stochastic arrivals and real-time service decisions. Foundational work emphasizes throughput and queue stability [12]–[14], while stochastic network optimization also studies tradeoffs between time-average objectives and backlog [15]. Our setting differs in its adversarial arrivals, convex state–action costs, and regret benchmark against a policy chosen in hindsight.

In many queueing applications, the natural retention coefficient is $\alpha = 1$ (no abandonment or leakage), so those systems are not direct instances of the $\alpha \in (0,1)$ model analyzed here. Section VII discusses the additional issues that arise when $\alpha \ge 1$.

Among other results, Khojastepour and Sabharwal [16] study a time-invariant robust scheduler representable as a linear filter of past arrivals. It is designed to minimize average transmit power subject to a maximum packet-delay guarantee, rather than updated online to minimize revealed general costs.

*d) Inventory management:* Yang et al. [17] study online procurement under inventory constraints. The controller observes the current demand and price, then procures supply while ensuring that demand is met; performance is measured by competitive ratio for aggregate procurement cost. Our controller instead chooses an outflow before observing the current inflow and cost, and permits convex state–action costs.

Hihat and Fermanian [18] consider an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions and study base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [17], the controller chooses supply and then observes demand through the transition, whereas our controller chooses an outflow and then observes incoming supply. Our scalar linear recursion and simplex policy class yield explicit feasibility and regret guarantees; the two frameworks target different dynamics, decisions, and policy classes.

*e) Water-reservoir management:* Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [19], [20]). Relative to our setting, this literature assumes stochastic inputs and emphasizes stationary or long-run performance quantities. It provides historical context for storage control under state constraints.

### F. Paper Organization

Section III formalizes the adversarial storage-control model, assumptions, and regret benchmark. Section IV introduces the SDAC policy class and summarizes its structural properties. Section V presents the online controller, Section VI develops regret guarantees, Section VII discusses possible extensions, and Section VIII concludes. Deferred proofs appear in the Appendix.

## II. Notation

We write $[z]_+ \triangleq \max\{z, 0\}$. For $v \in \mathbb{R}^n$, the $i$-th coordinate is $v^{(i)}$. We write $\mathbf{1}$ for the all-ones vector (dimension is clear from context). The Euclidean inner product is denoted by $\langle \cdot, \cdot \rangle$. We set $w_\tau \equiv 0$ for $\tau \le 0$. We write $\mathcal{A}$ for the online algorithm and $\pi$ for a fixed policy (also written $\pi(M)$ when parameterized by a vector $M$).

[1]"Disturbance-Action Control" is the standard name in the online-control literature. In this paper the underlying signal is the nonnegative arrival $w_t$; we retain the DAC/SDAC names for continuity with that literature.

## III. Model and Problem Setup

We consider a discrete-time process indexed by $t \in \{1, \dots, T\}$. The system state $x_t \in \mathbb{R}_{\geq 0}$ denotes the resource level at the beginning of time $t$. The initial state is given as

$$x_1 = 0. \tag{1}$$

Fix a retention coefficient $\alpha \in (0,1)$. At each time $t$, the controller chooses a control action $u_t \in \mathbb{R}_{\geq 0}$ subject to the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0,1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Every feasible trajectory therefore satisfies $0 \leq x_t \leq (1-\alpha)^{-1}$ and $0 \leq u_t \leq \alpha(1-\alpha)^{-1}$. Define the reachable state–action domain

$$\mathcal{D}_\alpha \triangleq \left\{ (x,u) : 0 \leq x \leq \frac{1}{1-\alpha},\ 0 \leq u \leq \alpha x \right\}. \tag{4}$$

After the control action is selected, a convex cost function

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

is revealed, and the incurred cost is $c_t(x_t, u_t)$. We assume each $c_t$ is convex in $(x,u)$ and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there exist constants $L_x, L_u > 0$, independent of $t$, such that for all $(x,u), (x',u') \in \mathcal{D}_\alpha$,

$$|c_t(x,u) - c_t(x',u')| \leq L_x|x - x'| + L_u|u - u'|. \tag{5}$$

We also assume access, at every $(x,u) \in \mathcal{D}_\alpha$, to a cost subgradient $s_t(x,u) = (s_{x,t}, s_{u,t}) \in \partial c_t(x,u)$ satisfying $|s_{x,t}| \leq L_x$ and $|s_{u,t}| \leq L_u$. This oracle condition holds, for example, when the Lipschitz property above extends to an open neighborhood of $\mathcal{D}_\alpha$.

The information pattern is adversarial and non-anticipatory. At time $t$, the environment may choose $(w_t, c_t)$ as a function of the history through time $t-1$, but not of the current action (or current randomization). The controller chooses $u_t$ without observing $(w_t, c_t)$; the pair is then revealed. Subject to this timing and the stated regularity conditions, the arrival and cost sequences may be arbitrary.

*a) Assumptions:*

1) **Retention coefficient:** $\alpha \in (0,1)$ is fixed and known (a fraction $1-\alpha$ of the state is lost each period).
2) **Arrival bounds:** $w_t \in [0,1]$ for all $t \in \{1, \dots, T\}$. In particular, arrivals are nonnegative resource increments, in contrast to signed process-noise disturbances common in classical control.
3) **Adversarial timing:** $(w_t, c_t)$ may depend on the history through time $t-1$ but not on the current action, and is revealed only after $u_t$ is chosen.
4) **Cost regularity:** each $c_t$ is convex in $(x,u)$, obeys (5) on $\mathcal{D}_\alpha$, and admits the bounded subgradient oracle described above.
5) **Feasible policy class:** both the online policy and the benchmark policies are required to satisfy (2) under (3).

We focus on $\alpha \in (0,1)$; the case $\alpha \geq 1$ is discussed in Section VII.

Let $\{(x_t^{\mathcal{A}}, u_t^{\mathcal{A}})\}_{t=1}^T$ denote the trajectory of an online algorithm $\mathcal{A}$. Define its cumulative cost by

$$J_T(\mathcal{A}) \triangleq \sum_{t=1}^{T} c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big). \tag{6}$$

For a fixed positive integer $H$, let $\pi$ range over the policy class $\mathrm{SDAC}_H$ (defined in Section IV). For each $\pi \in \mathrm{SDAC}_H$, let $\{(x_t^\pi, u_t^\pi)\}_{t=1}^T$ be the induced feasible trajectory under the same arrival sequence. The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \triangleq J_T(\mathcal{A}) - \min_{\pi \in \mathrm{SDAC}_H} \sum_{t=1}^{T} c_t\big(x_t^\pi, u_t^\pi\big). \tag{7}$$

While the online controller may update its SDAC parameters over time, the benchmark in (7) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that satisfies (2) for all $t \in \{1, \dots, T\}$ and achieves sublinear regret as defined in (7).

## IV. Simplex Disturbance-Action Control (SDAC)

### A. From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate losses and efficient online updates. In our scalar storage model, specializing the baseline feedback term to zero gives the representative form

$$u_t = \sum_{i=1}^{H} M^{(i)} w_{t-i}, \tag{8}$$

where $H \in \mathbb{N}$ is the memory horizon and $M \in \mathbb{R}^H$ is a linear filter. We allow nonpositive indices in such sums, with the convention $w_\tau \equiv 0$ for $\tau \leq 0$.

For our constrained model, an unrestricted filter is insufficient: its action can violate (2) by exceeding $\alpha x_t$ or becoming negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction yields fixed-policy feasibility while preserving convex surrogate losses and tractable online updates.

### B. SDAC Definition

Fix a memory horizon $H \in \mathbb{N}$. Let

$$\Delta_H \triangleq \left\{ M \in \mathbb{R}^H_{\geq 0} : \sum_{i=1}^{H} M^{(i)} \leq 1 \right\} \tag{9}$$

be the subprobability simplex. For any $M \in \Delta_H$, define the policy $\pi(M)$ by

$$u_t^{\pi(M)} \triangleq \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i}, \qquad w_\tau \equiv 0 \text{ for } \tau \leq 0. \tag{10}$$

For each time $t \in \{1, \dots, T\}$, let

$$\widetilde{w}_t \triangleq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \dots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H, \tag{11}$$

with the same convention $w_\tau \equiv 0$ for $\tau \le 0$, so that (10) can be written as

$$u_t^{\pi(M)} = \langle M, \widetilde{w}_t \rangle. \tag{12}$$

The induced state trajectory satisfies

$$x_{t+1}^{\pi(M)} = \alpha x_t^{\pi(M)} - u_t^{\pi(M)} + w_t, \qquad x_1^{\pi(M)} = 0. \tag{13}$$

The weights $\alpha^i$ align the arrival filter with the retention coefficient $\alpha$ in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in $M$, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on $M$.

Recall that we set $w_\tau \equiv 0$ for $\tau \le 0$, which is used in the base-case arguments below.

**Lemma 1** (State decomposition under SDAC)**.** *Fix $H \in \mathbb{N}$ and $M \in \Delta_H$, and write $W \triangleq \sum_{i=1}^H M^{(i)} \le 1$. Let $\pi(M)$ be defined by* (10) *and* (13)*. Define*

$$\psi_t^{(i)} \triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j} \quad (i = 1, \dots, H),$$

$$\rho_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j}.$$

*Then for all $t \in \{2, \dots, T\}$,*

$$x_t^{\pi(M)} = \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + \left(1 - W\right) \rho_t, \tag{14}$$

*and the coefficients satisfy $0 \le \psi_t^{(i)} \le \rho_t \le 1/(1-\alpha)$. In particular, the map $M \mapsto x_t^{\pi(M)}$ is affine.*

*Proof.* We first record a one-step recursion for $\rho_t$: isolating the $j = 1$ term in its definition and reindexing gives

$$\rho_{t+1} = \alpha \rho_t + w_t, \qquad \rho_1 = 0. \tag{15}$$

We prove (14) by induction on $t$.

*Base case* $t = 2$. Since $x_1^{\pi(M)} = 0$ and $w_{1-i} = 0$ for $i \ge 1$, (10) and (13) give

$$x_2^{\pi(M)} = \alpha \cdot 0 - 0 + w_1 = w_1.$$

Also $\psi_2^{(i)} = \rho_2 = w_1$ for every $i$, so the right side of (14) at $t = 2$ is

$$\sum_{i=1}^{H} M^{(i)} w_1 + (1-W) w_1 = W w_1 + (1-W) w_1 = w_1 = x_2^{\pi(M)}.$$

*Inductive step.* Assume (14) holds at time $t \in \{2, \dots, T-1\}$. By (13) and (10),

$$x_{t+1}^{\pi(M)} = \alpha x_t^{\pi(M)} - \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i} + w_t. \tag{16}$$

Substituting (14) for $x_t^{\pi(M)}$ in (16), and then distributing $\alpha$,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \alpha \Big( \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + (1-W)\rho_t \Big) \\ &\quad - \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H} M^{(i)} \Big( \alpha \psi_t^{(i)} - \alpha^i w_{t-i} \Big) + \alpha(1-W)\rho_t + w_t. \end{aligned} \tag{17}$$

By definition of $\psi_{t+1}^{(i)}$, for each $i$

$$\begin{aligned} \alpha \psi_t^{(i)} - \alpha^i w_{t-i} &= \sum_{j=1}^{i} \alpha^j w_{t-j} - \alpha^i w_{t-i} \\ &= \sum_{j=1}^{i-1} \alpha^j w_{t-j} \\ &= \psi_{t+1}^{(i)} - w_t. \end{aligned} \tag{18}$$

Substituting (18) into (17), and then summing over $i$,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \sum_{i=1}^{H} M^{(i)} \big( \psi_{t+1}^{(i)} - w_t \big) + \alpha(1-W)\rho_t + w_t \\ &= \sum_{i=1}^{H} M^{(i)} \psi_{t+1}^{(i)} - W w_t + \alpha(1-W)\rho_t + w_t. \end{aligned} \tag{19}$$

Regrouping the terms in $w_t$ and $\rho_t$, and then applying (15),

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \sum_{i=1}^{H} M^{(i)} \psi_{t+1}^{(i)} + (1-W)\big( w_t + \alpha \rho_t \big) \\ &= \sum_{i=1}^{H} M^{(i)} \psi_{t+1}^{(i)} + (1-W)\rho_{t+1}, \end{aligned}$$

which is (14) at time $t+1$.

*Bounds.* Each $\psi_t^{(i)}$ is the truncation to $j \le i$ of the nonnegative series defining $\rho_t$, so $0 \le \psi_t^{(i)} \le \rho_t$. Since $w_\tau \in [0, 1]$,

$$\rho_t = \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j} \le \sum_{j=1}^{\infty} \alpha^{j-1} = \frac{1}{1-\alpha}.$$

Affineness in $M$ holds because $\psi_t$ and $\rho_t$ do not depend on $M$. □

The first consequence of this decomposition is that feasibility is preserved within the class.

**Lemma 2** (Feasibility of SDAC policies)**.** *For every $M \in \Delta_H$, the trajectory induced by* (10) *and* (13) *satisfies*

$$0 \le u_t^{\pi(M)} \le \alpha x_t^{\pi(M)} \qquad \textit{for all } t \in \{1, \dots, T\}. \tag{20}$$

*Proof.* By (10),

$$u_t^{\pi(M)} = \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i}. \tag{21}$$

The right-hand side of (21) is nonnegative, so $u_t^{\pi(M)} \ge 0$.

*Upper bound,* $t = 1$. By (13), $x_1^{\pi(M)} = 0$. Specializing (21) and using $w_\tau \equiv 0$ for $\tau \le 0$,

$$\begin{aligned} u_1^{\pi(M)} &= \sum_{i=1}^{H} M^{(i)} \alpha^i w_{1-i} \\ &= 0 \\ &= \alpha x_1^{\pi(M)}. \end{aligned}$$

*Upper bound,* $t \ge 2$.

Let $W \triangleq \sum_{i=1}^{H} M^{(i)}$. Multiplying (14) by $\alpha$, and then distributing $\alpha$,

$$\begin{aligned} \alpha x_t^{\pi(M)} &= \alpha \Big( \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + (1-W)\rho_t \Big) \\ &= \alpha \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + \alpha(1-W)\rho_t. \end{aligned} \tag{22}$$

By (9) and Lemma 1, $W \le 1$ and $\rho_t \ge 0$, so $\alpha(1-W)\rho_t \ge 0$. Thus (22) yields

$$\alpha x_t^{\pi(M)} \ge \alpha \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)}. \tag{23}$$

Substituting the definition of $\psi_t^{(i)}$ into the right-hand side of (23), and then distributing $\alpha$,

$$\begin{aligned} \alpha \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} &= \alpha \sum_{i=1}^{H} M^{(i)} \left( \sum_{j=1}^{i} \alpha^{j-1} w_{t-j} \right) \\ &= \sum_{i=1}^{H} M^{(i)} \left( \sum_{j=1}^{i} \alpha^{j} w_{t-j} \right). \end{aligned} \tag{24}$$

Expanding the right-hand side of (24) as a double sum, and then changing the order of summation,

$$\begin{aligned} \sum_{i=1}^{H} M^{(i)} \left( \sum_{j=1}^{i} \alpha^{j} w_{t-j} \right) &= \sum_{i=1}^{H} \sum_{j=1}^{i} M^{(i)} \alpha^j w_{t-j} \\ &= \sum_{j=1}^{H} \sum_{i=j}^{H} M^{(i)} \alpha^j w_{t-j} \\ &= \sum_{j=1}^{H} \left( \sum_{i=j}^{H} M^{(i)} \right) \alpha^j w_{t-j}. \end{aligned} \tag{25}$$

Since $M^{(i)} \ge 0$ and $\alpha^j w_{t-j} \ge 0$, and then applying (21),

$$\begin{aligned} \sum_{j=1}^{H} \left( \sum_{i=j}^{H} M^{(i)} \right) \alpha^j w_{t-j} &\ge \sum_{j=1}^{H} M^{(j)} \alpha^j w_{t-j} \\ &= u_t^{\pi(M)}. \end{aligned} \tag{26}$$

Chaining (23)–(26) gives the upper bound in (20). □

The second consequence is that SDAC is compatible with online convex optimization: for fixed arrival sequences, $u_t^{\pi(M)}$ is linear in $M$ by (12), and $x_t^{\pi(M)}$ is affine in $M$ by Lemma 1. Since each $c_t$ is convex in $(x, u)$, this already shows that the induced cost is convex in $M$; the next lemma also quantifies its Lipschitz constant, which we use later to tune the online algorithm's step size.

**Lemma 3** (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(M) \triangleq c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big), \qquad M \in \Delta_H,$$

*and*

$$L \triangleq \alpha L_u + \frac{L_x}{1-\alpha}. \tag{27}$$

*Then $\ell_t$ is convex on $\Delta_H$, and for all $M, M' \in \Delta_H$,*

$$|\ell_t(M) - \ell_t(M')| \le L \|M - M'\|_1.$$

*Moreover, at every $M \in \Delta_H$, one can select a subgradient $g \in \partial \ell_t(M)$ satisfying $\|g\|_\infty \le L$.*

*Proof.* For $t = 1$, (13) and (10) give $x_1^{\pi(M)} = u_1^{\pi(M)} = 0$, so $\ell_1$ is constant and the claims hold with $g = 0$. Fix $t \ge 2$, and write $W \triangleq \sum_{i=1}^{H} M^{(i)}$. By (14), and then substituting $W = \sum_{i=1}^{H} M^{(i)}$,

$$\begin{aligned} x_t^{\pi(M)} &= \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + (1-W)\rho_t \\ &= \sum_{i=1}^{H} M^{(i)} \big(\psi_t^{(i)} - \rho_t\big) + \rho_t. \end{aligned} \tag{28}$$

The right-hand side of (28) is affine in $M$, and (10) is linear in $M$. Since $c_t$ is convex, $\ell_t$ is convex.

Subtracting (28) at $M'$ from (28) at $M$, and (10) at $M'$ from (10) at $M$,

$$\begin{aligned} x_t^{\pi(M)} - x_t^{\pi(M')} &= \sum_{i=1}^{H} (M^{(i)} - M'^{(i)})(\psi_t^{(i)} - \rho_t), \\ u_t^{\pi(M)} - u_t^{\pi(M')} &= \sum_{i=1}^{H} (M^{(i)} - M'^{(i)}) \alpha^i w_{t-i}. \end{aligned} \tag{29}$$

Substituting (29) into (5),

$$\begin{aligned} &\big|\ell_t(M) - \ell_t(M')\big| \\ &\quad \le L_x \big| x_t^{\pi(M)} - x_t^{\pi(M')} \big| \\ &\qquad + L_u \big| u_t^{\pi(M)} - u_t^{\pi(M')} \big| \\ &\quad = L_x \left| \sum_{i=1}^{H} (M^{(i)} - M'^{(i)})(\psi_t^{(i)} - \rho_t) \right| \\ &\qquad + L_u \left| \sum_{i=1}^{H} (M^{(i)} - M'^{(i)}) \alpha^i w_{t-i} \right|. \end{aligned} \tag{30}$$

Applying Hölder's inequality to (30),

$$\begin{aligned} &\big|\ell_t(M) - \ell_t(M')\big| \\ &\le \big( L_x \max_i |\psi_t^{(i)} - \rho_t| + L_u \max_i \alpha^i w_{t-i} \big) \|M - M'\|_1. \end{aligned} \tag{31}$$

By Lemma 1, $0 \le \psi_t^{(i)} \le \rho_t \le 1/(1-\alpha)$, and $w_{t-i} \in [0, 1]$ with $i \ge 1$, so

$$\begin{aligned} \max_i \big|\psi_t^{(i)} - \rho_t\big| &\le \frac{1}{1-\alpha}, \\ \max_i \alpha^i w_{t-i} &\le \alpha. \end{aligned} \tag{32}$$

Substituting (32) into (31), and then (27),

$$\begin{aligned}\big|\ell_t(M) - \ell_t(M')\big| &\le \Big(\frac{L_x}{1-\alpha} + \alpha L_u\Big)\|M - M'\|_1 \\ &= L\|M - M'\|_1.\end{aligned}$$

Let $(s_x, s_u) \in \partial c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big)$ with $|s_x| \le L_x$ and $|s_u| \le L_u$, and set

$$g^{(i)} \triangleq s_x\big(\psi_t^{(i)} - \rho_t\big) + s_u \alpha^i w_{t-i}. \tag{33}$$

Affine composition with (28) and (10) gives $g \in \partial \ell_t(M)$. Using (32),

$$\begin{aligned}\big|g^{(i)}\big| &\le |s_x|\,\big|\psi_t^{(i)} - \rho_t\big| + |s_u|\,\alpha^i w_{t-i} \\ &\le \frac{L_x}{1-\alpha} + \alpha L_u \\ &= L.\end{aligned} \tag{34}$$

Thus $\|g\|_\infty \le L$. □

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

### C. Expressiveness: Proportional State-Feedback Approximation

A standard policy class in online control is the family of linear state-feedback policies $u_t = -Kx_t$, and a foundational property of Disturbance-Action Control is that the DAC class *approximates* any strongly stable linear policy, with approximation error decaying geometrically in the memory horizon $H$ [1], [4].

Under our sign convention, positive $u_t$ drains storage, so proportional feedback is written $u_t = qx_t$. In our scalar model, the feasible proportional state-feedback policies are exactly those with gain $q \in [0, \alpha]$: since $x_t \ge 0$, the feasibility constraint $0 \le u_t \le \alpha x_t$ in (2) holds if and only if $q \in [0, \alpha]$. The next lemma gives an explicit approximation within the feasible class $\mathrm{SDAC}_H$; the two endpoint policies are represented exactly.

**Lemma 4** (SDAC approximation of proportional state feedback)**.** *Fix $q \in [0, \alpha]$, set $\beta \triangleq \alpha - q$, and let $\pi_q^{\mathrm{lin}}$ denote the feasible linear policy*

$$\begin{aligned}u_t^{\mathrm{lin}} = q\, x_t^{\mathrm{lin}}, &\qquad x_{t+1}^{\mathrm{lin}} = \beta\, x_t^{\mathrm{lin}} + w_t, \\ t \in \{1, \dots, T\}, &\qquad x_1^{\mathrm{lin}} = 0.\end{aligned}$$

*Define $M_q^\star \in \mathbb{R}^H$ by*

$$M_q^{\star(i)} \triangleq \frac{q\,\beta^{i-1}}{\alpha^i}, \qquad i = 1, \dots, H.$$

*Then $M_q^\star \in \Delta_H$, and if each $c_t$ satisfies (5), then for every $\{w_t\} \subset [0, 1]$,*

$$\begin{aligned}&\Big|\sum_{t=1}^T c_t\big(x_t^{\pi(M_q^\star)}, u_t^{\pi(M_q^\star)}\big) - \sum_{t=1}^T c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big)\Big| \\ &\qquad \le \Big(L_u + \frac{L_x}{1-\alpha}\Big)\frac{q\,\beta^H}{1-\beta}\,T.\end{aligned} \tag{35}$$

*The proof is deferred to Appendix A.*

Theorem 1 depends on the memory horizon only through $\sqrt{\log(H+1)}$. Combining that bound with Lemma 4 therefore yields a no-regret guarantee against the best feasible proportional state-feedback policy; Corollary 1 makes this statement precise.

## V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by online mirror descent (OMD) with unnormalized negentropy (equivalently, the exponentiated gradient method) and a fixed step size $\eta > 0$ over the sub-probability simplex $\Delta_H \subset \mathbb{R}^H$ [21].

For each time $t \in \{1, \dots, T\}$, let $M_t \in \Delta_H$ be the online parameter. We initialize at

$$M_1^{(i)} \triangleq \frac{1}{H+1}, \qquad i = 1, \dots, H \tag{36}$$

so that the maximum initial divergence $\sup_{M \in \Delta_H} D_\Psi(M \| M_1)$ is at most $\log(H+1)$ (see the proof of Theorem 1).

*a) Control-action selection:* The controller first forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle M_t, \widetilde{w}_t \rangle, \tag{37}$$

and clips it to the admissible interval $[0, \alpha x_t^{\mathcal{A}}]$:

$$u_t^{\mathcal{A}} \triangleq \min\{\widehat{u}_t,\, \alpha x_t^{\mathcal{A}}\}. \tag{38}$$

The state update is

$$x_{t+1}^{\mathcal{A}} = \alpha x_t^{\mathcal{A}} - u_t^{\mathcal{A}} + w_t, \qquad x_1^{\mathcal{A}} = 0. \tag{39}$$

By construction, $\widehat{u}_t \ge 0$ and $u_t^{\mathcal{A}} \in [0, \alpha x_t^{\mathcal{A}}]$; with $x_1^{\mathcal{A}} = 0$ and $w_t \in [0, 1]$, induction on (39) yields $x_t^{\mathcal{A}} \ge 0$ for all $t \in \{1, \dots, T\}$, so the online trajectory satisfies (2).

*b) Surrogate cost and OMD update:* After $(w_t, c_t)$ is revealed, define the surrogate cost by

$$\ell_t(M) \triangleq c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big), \qquad M \in \Delta_H, \tag{40}$$

which is convex and Lipschitz by Lemma 3. Select a subgradient $g_t \in \partial \ell_t(M_t)$ satisfying $\|g_t\|_\infty \le L$, whose existence and construction are given in that lemma. Let

$$\Psi(M) \triangleq \sum_{i=1}^H \big(M^{(i)} \log M^{(i)} - M^{(i)}\big), \qquad M \in \mathbb{R}^H_{\ge 0}, \tag{41}$$

denote the unnormalized negentropy (with the convention $0 \log 0 \triangleq 0$), and let

$$D_\Psi(M \| V) \triangleq \sum_{i=1}^H \Big(M^{(i)} \log \tfrac{M^{(i)}}{V^{(i)}} - M^{(i)} + V^{(i)}\Big)$$

be its Bregman divergence for $V \in \mathbb{R}^H_{>0}$, the generalized Kullback–Leibler divergence.

We define the OMD update

$$M_{t+1} \triangleq \operatorname*{arg\,min}_{M \in \Delta_H} \Big\{\eta \langle g_t, M\rangle + D_\Psi(M \| M_t)\Big\}. \tag{42}$$

The regret analysis in Section VI works directly with this optimization-based definition.

*c) Closed-form solution:* Although (42) is a constrained optimization problem, it is easy to solve analytically because $\Psi$ is separable and the objective is linear in $g_t$. As shown in Lemma 6, the minimizer in (42) is exactly the result of two elementary steps: a multiplicative step,

$$\widetilde{M}_{t+1}^{(i)} \triangleq M_t^{(i)} \exp\bigl(-\eta g_t^{(i)}\bigr), \qquad i = 1, \dots, H, \tag{43}$$

followed by the Bregman projection onto $\Delta_H$ given by (44) (the scaling is nontrivial only when $S_{t+1} > 1$):

$$M_{t+1}^{(i)} \triangleq \frac{\widetilde{M}_{t+1}^{(i)}}{\max\{1, S_{t+1}\}}, \quad S_{t+1} \triangleq \sum_{j=1}^{H} \widetilde{M}_{t+1}^{(j)}, \tag{44}$$

for $i = 1, \dots, H$. This is the form we implement in practice.

## VI. Regret Analysis

The following theorem states that the online controller achieves sublinear regret against every fixed SDAC policy $\pi(M)$, where $M$ is any parameter in $\Delta_H$. The regret proof has two parts: a standard online convex optimization bound for the surrogate costs $\ell_t$, and a stability bound showing that the action-clipped online trajectory stays close to the SDAC policy trajectory driven by the same parameters. Some technical lemmas required for the proof are deferred to Appendix A.

**Theorem 1** (Regret bound)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section III. Then the OMD controller with action clipping from Section V, initialized as in* (36) *and using step size* $\eta > 0$*, satisfies*

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \le \frac{\log(H+1)}{\eta} + \eta L^2 T\Bigl(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Bigr),$$
$$L \triangleq \alpha L_u + \tfrac{L_x}{1-\alpha}.$$

*In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2 T\bigl(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\bigr)}}$$

*gives*

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \le 2L\sqrt{\Bigl(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Bigr) T \log(H+1)}.$$

Combining Theorem 1 with Lemma 4 yields a uniform guarantee against the best feasible proportional state-feedback policy.

**Corollary 1** (No-regret against proportional state feedback)**.** *Set*

$$H_T \triangleq \max\Bigl\{1, \Bigl\lceil \frac{\log T}{\log(1/\alpha)} \Bigr\rceil\Bigr\}.$$

*Run the OMD controller with action clipping using memory horizon* $H = H_T$ *and the step size* $\eta^*$ *of Theorem 1. For each* $q \in [0, \alpha]$*, let* $\bigl(x_t^{(q)}, u_t^{(q)}\bigr)$ *denote the trajectory of the proportional policy* $u_t^{(q)} = q x_t^{(q)}$*. Define*

$$\mathrm{Reg}_{\mathrm{lin}}(T) \triangleq \sum_{t=1}^{T} c_t\bigl(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\bigr) - \min_{q\in[0,\alpha]} \sum_{t=1}^{T} c_t\bigl(x_t^{(q)}, u_t^{(q)}\bigr).$$

*Then*

$$\mathrm{Reg}_{\mathrm{lin}}(T) \le 2L\sqrt{\Bigl(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Bigr) T \log(H_T+1)} + \Bigl(L_u + \frac{L_x}{1-\alpha}\Bigr)\frac{\alpha^{H_T+1}}{1-\alpha}\, T,$$

*where* $L = \alpha L_u + \frac{L_x}{1-\alpha}$*. Since* $\alpha^{H_T} \le 1/T$*, the second term is* $O(1)$*, while* $\log(H_T+1) = O(\log\log T)$ *as* $T \to \infty$*. Consequently,*

$$\mathrm{Reg}_{\mathrm{lin}}(T) = O\Bigl(\sqrt{T \log\log T}\Bigr).$$

*Thus* $\mathrm{Reg}_{\mathrm{lin}}(T) = o(T)$ *against the best proportional state-feedback policy with* $q \in [0, \alpha]$*.*

*Proof.* Fix any $q \in [0, \alpha]$ and set $\beta = \alpha - q$. Theorem 1, followed by Lemma 4, gives

$$\sum_{t=1}^{T} c_t\bigl(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\bigr) - \sum_{t=1}^{T} c_t\bigl(x_t^{(q)}, u_t^{(q)}\bigr)$$
$$\le 2L\sqrt{\Bigl(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Bigr) T \log(H_T+1)} + \Bigl(L_u + \frac{L_x}{1-\alpha}\Bigr)\frac{q\,\beta^{H_T}}{1-\beta}\, T.$$

Because $q \le \alpha$, $\beta \le \alpha$, and $1 - \beta \ge 1 - \alpha$,

$$\frac{q\,\beta^{H_T}}{1-\beta} \le \frac{\alpha^{H_T+1}}{1-\alpha}.$$

This uniform bound holds for every $q \in [0, \alpha]$; minimizing the comparator cost over $q$ proves the stated inequality. Finally, the definition of $H_T$ gives $\alpha^{H_T} \le 1/T$ and $H_T = \Theta(\log T)$ as $T \to \infty$, which establishes the asymptotic rate. ☐

## VII. Extensions

We briefly discuss three possible extensions. Establishing their full regret bounds is left for future work.

*a) Retention* $\alpha \ge 1$*.:* For $\alpha \ge 1$, choose a fixed baseline drain $K x_t$ such that $\bar{\alpha} \triangleq \alpha - K \in (0, 1)$, and write $u_t = K x_t + v_t$. The residual dynamics become $x_{t+1} = \bar{\alpha} x_t - v_t + w_t$. An SDAC filter for $v_t$ should therefore use the powers $\bar{\alpha}^i$, not $\alpha^i$. If $0 \le v_t \le \bar{\alpha} x_t$, then $0 \le u_t \le (K + \bar{\alpha}) x_t = \alpha x_t$. This reduction suggests an extension of the present analysis with transformed cost and Lipschitz constants; we do not pursue the details here.

*b) Vector action / multi-task allocation.:* Consider a single storage state $x_t$ with a vector action $u_t \in \mathbb{R}^n_{\ge 0}$ that allocates resources across $n$ tasks, subject to $\mathbf{1}^\top u_t \le \alpha x_t$, so that the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \mathbf{1}^\top u_t + w_t$, and the cost $c_t(x_t, u_t)$ is convex in $(x, u)$. The fixed-policy SDAC construction lifts by replacing the filter vector $M \in \Delta_H$ with a matrix $M \in \mathbb{R}^{n\times H}_{\ge 0}$ whose entries sum to at most one (a matrix simplex), with each row corresponding to a task and each column to a memory lag. OMD can then run over this matrix simplex, with radial action clipping when the nominal total allocation exceeds $\alpha x_t$.

*c) Multiple storages with diagonal dynamics.:* Suppose the state is $x_t \in \mathbb{R}^d_{\geq 0}$ and evolves as $x_{t+1} = Ax_t - u_t + w_t$ with $A = \mathrm{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_i \in (0,1)$, arrivals $w_t \in [0,1]^d$, and per-coordinate constraints $0 \leq u_t^{(i)} \leq \alpha_i x_t^{(i)}$. The dynamics and feasible set then decouple by coordinate, and the natural SDAC class is a product of simplices. For a jointly convex cost, OMD operates jointly on this product set; the optimization separates by coordinate only when the cost is coordinate-separable. Cross-coupled storage systems (non-diagonal $A$) require additional analysis.

These constructions retain the basic simplex/OMD structure, but their complete feasibility and regret analyses require separate treatment.

## VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. Every fixed SDAC policy is feasible by construction. For the time-varying online controller, a Bregman projection maintains the SDAC parameter constraint, while scalar action clipping maintains feasibility of the actual trajectory. The resulting controller achieves $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. The SDAC approximation result further gives $O\big(\sqrt{T\log\log T}\big)$ regret against the best feasible proportional state-feedback policy when $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time. Section VII discusses possible extensions to $\alpha \geq 1$, vector allocation, and diagonal multi-storage systems.

## Appendix A
## Deferred Proofs

This appendix collects the deferred proof of the proportional-state-feedback approximation guarantee from Section IV, together with the OMD and stability lemmas leading up to the regret bound stated in Section VI.

Recall that we set $w_\tau \equiv 0$ for $\tau \leq 0$, which is used in several base-case arguments below.

We first prove the linear-policy approximation guarantee stated in Section IV-C.

*Proof of Lemma 4.* Fix $q \in [0,\alpha]$, set $\beta \triangleq \alpha - q \in [0,\alpha]$, and recall $w_\tau \equiv 0$ for $\tau \leq 0$.

*Feasibility and closed form.* Substituting $u_t^{\mathrm{lin}} = q\,x_t^{\mathrm{lin}}$ into (3),

$$\begin{aligned} x_{t+1}^{\mathrm{lin}} &= \alpha x_t^{\mathrm{lin}} - q x_t^{\mathrm{lin}} + w_t \\ &= \beta x_t^{\mathrm{lin}} + w_t, \end{aligned} \tag{45}$$

with $x_1^{\mathrm{lin}} = 0$ by (1). Since $\beta \geq 0$ and $w_t \geq 0$, if $x_t^{\mathrm{lin}} \geq 0$ then

$$x_{t+1}^{\mathrm{lin}} = \beta x_t^{\mathrm{lin}} + w_t \geq 0.$$

Thus $x_t^{\mathrm{lin}} \geq 0$ for all $t$, and $q \in [0,\alpha]$ gives

$$0 \leq u_t^{\mathrm{lin}} = q\,x_t^{\mathrm{lin}} \leq \alpha x_t^{\mathrm{lin}},$$

which is (2). Unrolling (45) from $x_1^{\mathrm{lin}} = 0$ yields, for all $t \in \{1,\ldots,T\}$,

$$\begin{aligned} x_t^{\mathrm{lin}} &= \sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}, \\ u_t^{\mathrm{lin}} &= q\sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}. \end{aligned} \tag{46}$$

*Simplex membership.* Each $M_q^{\star(i)} = q\beta^{i-1}/\alpha^i \geq 0$. If $q = 0$, then $M_q^\star = 0 \in \Delta_H$. If $q > 0$,

$$\begin{aligned} 1 - \frac{\beta}{\alpha} &= \frac{\alpha-\beta}{\alpha} \\ &= \frac{q}{\alpha}. \end{aligned} \tag{47}$$

Hence

$$\begin{aligned} \sum_{i=1}^H M_q^{\star(i)} &= \sum_{i=1}^H \frac{q\,\beta^{i-1}}{\alpha^i} \\ &= \frac{q}{\alpha}\sum_{i=1}^H \Big(\frac{\beta}{\alpha}\Big)^{i-1} \\ &= \frac{q}{\alpha}\cdot\frac{1-(\beta/\alpha)^H}{1-\beta/\alpha} \\ &= \frac{q}{\alpha}\cdot\frac{\alpha}{q}\big(1-(\beta/\alpha)^H\big) \\ &= 1-(\beta/\alpha)^H \\ &\leq 1, \end{aligned}$$

where the fourth equality uses (47) and the inequality uses $0 \leq \beta/\alpha < 1$. Thus $M_q^\star \in \Delta_H$.

*Action approximation.* By definition, $M_q^{\star(i)}\alpha^i = q\beta^{i-1}$. Substituting into (10),

$$\begin{aligned} u_t^{\pi(M_q^\star)} &= \sum_{i=1}^H M_q^{\star(i)}\alpha^i w_{t-i} \\ &= q\sum_{i=1}^H \beta^{i-1} w_{t-i}. \end{aligned} \tag{48}$$

Using $w_{t-i} = 0$ for $i \geq t$ in (48), and then subtracting from (46),

$$\begin{aligned} &u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)} \\ &\quad= q\sum_{j=1}^{t-1}\beta^{j-1}w_{t-j} \\ &\qquad - q\sum_{i=1}^{\min\{H,t-1\}}\beta^{i-1}w_{t-i} \\ &\quad= q\sum_{j=H+1}^{t-1}\beta^{j-1}w_{t-j}. \end{aligned} \tag{49}$$

The right-hand side of (49) is nonnegative. Since $w_{t-j} \in [0,1]$,

$$\begin{aligned} u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)} &\leq q\sum_{j=H+1}^{\infty}\beta^{j-1} \\ &= \frac{q\,\beta^H}{1-\beta}. \end{aligned} \tag{50}$$

*State approximation.* Let $\delta_t \triangleq x_t^{\pi(M_q^\star)} - x_t^{\mathrm{lin}}$ and $e_t \triangleq u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)}$. By (49) and (50),

$$0 \le e_t \le \frac{q\,\beta^H}{1-\beta}. \tag{51}$$

Subtracting the two copies of (3),

$$\begin{aligned}\delta_{t+1} &= \big(\alpha x_t^{\pi(M_q^\star)} - u_t^{\pi(M_q^\star)} + w_t\big)\\ &\quad - \big(\alpha x_t^{\mathrm{lin}} - u_t^{\mathrm{lin}} + w_t\big)\\ &= \alpha\delta_t + e_t,\end{aligned} \tag{52}$$

with $\delta_1 = 0$. Unrolling (52) gives

$$\delta_t = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s. \tag{53}$$

By (51), the right-hand side of (53) is nonnegative. Substituting the upper bound of (51) into (53),

$$\begin{aligned}\delta_t &\le \frac{q\,\beta^H}{1-\beta}\sum_{s=1}^{t-1}\alpha^{t-1-s}\\ &\le \frac{q\,\beta^H}{(1-\beta)(1-\alpha)}.\end{aligned} \tag{54}$$

*Cost approximation.* By (5), and then substituting (54) and (51),

$$\begin{aligned}&\big|c_t\big(x_t^{\pi(M_q^\star)}, u_t^{\pi(M_q^\star)}\big) - c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big)\big|\\ &\quad\le L_x|\delta_t| + L_u|e_t|\\ &\quad\le L_x \cdot \frac{q\,\beta^H}{(1-\beta)(1-\alpha)} + L_u \cdot \frac{q\,\beta^H}{1-\beta}\\ &\quad= \Big(L_u + \frac{L_x}{1-\alpha}\Big)\frac{q\,\beta^H}{1-\beta}.\end{aligned} \tag{55}$$

Summing (55) over $t = 1, \dots, T$ and applying the triangle inequality yields (35). □

The remaining lemmas establish the OMD and stability estimates used in the proof of Theorem 1; they rely on the state-decomposition, feasibility, and surrogate-cost lemmas already established in Section IV (Lemmas 1, 2, and 3).

**Lemma 5** (Bregman projection onto $\Delta_H$)**.** *Let $\Psi$ and $D_\Psi$ be the unnormalized negentropy and its Bregman divergence from* (41)*. Fix $\widetilde{M} \in \mathbb{R}^H_{>0}$ and set $S \triangleq \sum_{i=1}^H \widetilde{M}^{(i)}$. Then the Bregman projection of $\widetilde{M}$ onto $\Delta_H = \{M \in \mathbb{R}^H_{\ge 0} : \sum_i M^{(i)} \le 1\}$,*

$$P(\widetilde{M}) \triangleq \operatorname*{arg\,min}_{M\in\Delta_H} D_\Psi(M\|\widetilde{M}),$$

*is given coordinatewise by*

$$P(\widetilde{M})^{(i)} = \frac{\widetilde{M}^{(i)}}{\max\{1, S\}}.$$

*Proof.* Since $\widetilde{M}$ is fixed, the terms $+\widetilde{M}^{(i)}$ are constant, so $P(\widetilde{M})$ minimizes

$$F(M) \triangleq \sum_{i=1}^H \Big(M^{(i)} \log \tfrac{M^{(i)}}{\widetilde{M}^{(i)}} - M^{(i)}\Big)$$

over $\Delta_H$. The function $F$ is strictly convex on $\mathbb{R}^H_{\ge 0}$ and $\Delta_H$ is closed and convex, so the minimizer is unique. On $\mathbb{R}^H_{>0}$, its partial derivatives are $\partial F/\partial M^{(i)} = \log\big(M^{(i)}/\widetilde{M}^{(i)}\big)$.

*Case* $S \le 1$. Here $\widetilde{M} \in \Delta_H$. Setting $\partial F/\partial M^{(i)} = 0$ gives the unconstrained minimizer $M = \widetilde{M}$ over $\mathbb{R}^H_{\ge 0}$, which is feasible; hence $P(\widetilde{M}) = \widetilde{M}$.

*Case* $S > 1$. Here $\widetilde{M} \notin \Delta_H$, so the constraint $\sum_i M^{(i)} \le 1$ must be active at the optimum. As $\widetilde{M}^{(i)} > 0$, the optimal coordinates are strictly positive and the nonnegativity constraints are inactive; introducing a multiplier $\lambda$ for $\sum_i M^{(i)} = 1$, KKT stationarity reads

$$\log\big(M^{(i)}/\widetilde{M}^{(i)}\big) + \lambda = 0 \quad\Longrightarrow\quad M^{(i)} = \widetilde{M}^{(i)} e^{-\lambda}.$$

The active constraint $\sum_i M^{(i)} = 1$ forces $e^{-\lambda} = 1/S$, so $M^{(i)} = \widetilde{M}^{(i)}/S$; moreover $\lambda = \log S > 0$, so dual feasibility holds. Equivalently, $P(\widetilde{M})^{(i)} = \widetilde{M}^{(i)}/\max\{1, S\}$ in both cases. □

**Lemma 6** (Closed-form solution of the OMD update)**.** *Fix $M_t \in \Delta_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $g_t \in \mathbb{R}^H$, and let $\widetilde{M}_{t+1}$ be defined coordinatewise by* (43)*, i.e. $\widetilde{M}_{t+1}^{(i)} = M_t^{(i)} e^{-\eta g_t^{(i)}}$. Then*

$$\begin{aligned}&\operatorname*{arg\,min}_{M\in\Delta_H}\Big\{\eta\langle g_t, M\rangle + D_\Psi(M\|M_t)\Big\}\\ &\qquad= \operatorname*{arg\,min}_{M\in\Delta_H} D_\Psi(M\|\widetilde{M}_{t+1}) = P(\widetilde{M}_{t+1}),\end{aligned}$$

*where $P(\cdot)$ is the Bregman projection of Lemma 5. Consequently, the OMD update* (42) *coincides exactly with the closed-form OMD update* (43)–(44)*, and in particular $M_{t+1} \in \Delta_H \cap \mathbb{R}^H_{>0}$. Thus the positive initialization* (36) *keeps every algorithm iterate strictly positive.*

*Proof.* For any $M \in \mathbb{R}^H_{\ge 0}$, expand the objective using the definitions of $D_\Psi$ from (41) and of $\widetilde{M}_{t+1}$:

$$\begin{aligned}&\eta\langle g_t, M\rangle + D_\Psi(M\|M_t)\\ &\quad= \sum_{i=1}^H\Big(\eta g_t^{(i)} M^{(i)} + M^{(i)}\log\tfrac{M^{(i)}}{M_t^{(i)}} - M^{(i)} + M_t^{(i)}\Big)\\ &\quad= \sum_{i=1}^H\Big(M^{(i)}\log\tfrac{M^{(i)}}{M_t^{(i)}e^{-\eta g_t^{(i)}}} - M^{(i)} + M_t^{(i)}\Big)\\ &\quad= \sum_{i=1}^H\Big(M^{(i)}\log\tfrac{M^{(i)}}{\widetilde{M}_{t+1}^{(i)}} - M^{(i)} + \widetilde{M}_{t+1}^{(i)}\Big)\\ &\qquad+ \sum_{i=1}^H\big(M_t^{(i)} - \widetilde{M}_{t+1}^{(i)}\big)\\ &\quad= D_\Psi(M\|\widetilde{M}_{t+1}) + \sum_{i=1}^H\big(M_t^{(i)} - \widetilde{M}_{t+1}^{(i)}\big).\end{aligned}$$

The second sum does not depend on $M$, so minimizing the left-hand side over $M \in \Delta_H$ is equivalent to minimizing $D_\Psi(M\|\widetilde{M}_{t+1})$ over $M \in \Delta_H$, which by definition is the Bregman projection $P(\widetilde{M}_{t+1})$. Lemma 5 identifies $P(\widetilde{M}_{t+1})$ coordinatewise as $\widetilde{M}_{t+1}^{(i)}/\max\{1, S_{t+1}\}$ with $S_{t+1} = \sum_j \widetilde{M}_{t+1}^{(j)}$, which is exactly (44). □

**Lemma 7** (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $g_t \in \partial \ell_t(M_t)$,*

$$\|M_{t+1} - M_t\|_1 \leq \eta \, \|g_t\|_\infty .$$

*In particular, if $\|g_t\|_\infty \leq L$, then $\|M_{t+1} - M_t\|_1 \leq \eta L$.*

*Proof.* Let $\Psi$ and $D_\Psi$ be as in (41). By definition (42), $M_{t+1}$ minimizes the convex function $F(M) \triangleq \eta\langle g_t, M\rangle + D_\Psi(M\|M_t)$ over the convex set $\Delta_H$, and $F$ is differentiable on $\mathbb{R}^H_{>0}$ with $\nabla F(M) = \eta g_t + \nabla\Psi(M) - \nabla\Psi(M_t)$. The first-order optimality condition for constrained convex minimization therefore gives, for all $M \in \Delta_H$,

$$\left\langle g_t + \tfrac{1}{\eta}\big(\nabla\Psi(M_{t+1}) - \nabla\Psi(M_t)\big), M - M_{t+1} \right\rangle \geq 0.$$

Taking $M = M_t$ and rearranging yields

$$\langle g_t, M_t - M_{t+1}\rangle \geq \tfrac{1}{\eta}\left\langle \nabla\Psi(M_{t+1}) - \nabla\Psi(M_t),\, M_{t+1} - M_t \right\rangle.$$

By the Bregman identity,

$$\begin{aligned}&\left\langle \nabla\Psi(M_{t+1}) - \nabla\Psi(M_t),\, M_{t+1} - M_t \right\rangle \\ &\qquad = D_\Psi(M_{t+1}\|M_t) + D_\Psi(M_t\|M_{t+1}).\end{aligned}$$

Hence, using Hölder's inequality,

$$\begin{aligned}&D_\Psi(M_{t+1}\|M_t) + D_\Psi(M_t\|M_{t+1}) \\ &\qquad \leq \eta\,\|g_t\|_\infty\,\|M_{t+1} - M_t\|_1 .\end{aligned}$$

Since both iterates lie in $\Delta_H$, the unnormalized negentropy is 1-strongly convex with respect to $\|\cdot\|_1$ on this set: the Hessian $\nabla^2\Psi(M) = \mathrm{diag}(1/M^{(i)})$ satisfies $v^\top \nabla^2\Psi(M)\, v = \sum_i (v^{(i)})^2/M^{(i)} \geq \|v\|_1^2/(\sum_i M^{(i)}) \geq \|v\|_1^2$ whenever $\sum_i M^{(i)} \leq 1$ (Cauchy–Schwarz), so the usual Pinsker-type bound for the Bregman divergence extends to the subprobability simplex $\Delta_H$,

$$\begin{aligned}D_\Psi(M_{t+1}\|M_t) &\geq \tfrac{1}{2}\|M_{t+1} - M_t\|_1^2, \\ D_\Psi(M_t\|M_{t+1}) &\geq \tfrac{1}{2}\|M_{t+1} - M_t\|_1^2,\end{aligned}$$

and the left-hand side is at least $\|M_{t+1} - M_t\|_1^2$. Therefore

$$\|M_{t+1} - M_t\|_1^2 \leq \eta\,\|g_t\|_\infty\,\|M_{t+1} - M_t\|_1,$$

which implies $\|M_{t+1} - M_t\|_1 \leq \eta\|g_t\|_\infty$. □

**Lemma 8** (Cumulative movement)**.** *For every $t \in \{2, \ldots, T\}$ and every integer $i$ with $1 \leq i \leq t-1$,*

$$\|M_{t-i} - M_t\|_1 \leq i\,\eta L.$$

*Proof.* The range $1 \leq i \leq t-1$ ensures that every index $t-j$ for $j = 1, \ldots, i$ satisfies $t - j \geq 1$. By the triangle inequality and Lemma 7,

$$\|M_{t-i} - M_t\|_1 \leq \sum_{j=1}^{i} \|M_{t-j} - M_{t-j+1}\|_1 \leq i\,\eta L.$$

□

**Lemma 9** (Control-action bounds relative to a counterfactual trajectory)**.** *For all $t \in \{1, \ldots, T\}$,*

$$\begin{aligned}u_t^{\mathcal{A}} &\geq \langle M_t, \widetilde{w}_t\rangle - \alpha\,[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}]_+, \\ u_t^{\mathcal{A}} &\leq \langle M_t, \widetilde{w}_t\rangle.\end{aligned}$$

*Proof.* Recall from (37)–(38) that

$$\widehat{u}_t = \langle M_t, \widetilde{w}_t\rangle, \qquad u_t^{\mathcal{A}} = \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\}.$$

Therefore,

$$u_t^{\mathcal{A}} = \widehat{u}_t - \big[\widehat{u}_t - \alpha x_t^{\mathcal{A}}\big]_+ = \langle M_t, \widetilde{w}_t\rangle - \big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+.$$

This identity gives the upper bound immediately, since the subtracted term is nonnegative.

For the lower bound, Lemma 2 applied to the counterfactual fixed policy $\pi(M_t)$ yields, using (12),

$$u_t^{\pi(M_t)} = \langle M_t, \widetilde{w}_t\rangle \leq \alpha x_t^{\pi(M_t)}.$$

Subtracting $\alpha x_t^{\mathcal{A}}$ from both sides and applying monotonicity of $[\cdot]_+$ gives

$$\begin{aligned}&\big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+ \\ &\qquad \leq \big[\alpha x_t^{\pi(M_t)} - \alpha x_t^{\mathcal{A}}\big]_+ = \alpha\big[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}\big]_+.\end{aligned}$$

Substituting this bound into the expression above for $u_t^{\mathcal{A}}$ yields the stated lower bound. □

**Lemma 10** (State-gap recursion)**.** *Define the state gap*

$$\Gamma_t^M \triangleq x_t^{\pi(M)} - x_t^{\mathcal{A}}. \tag{56}$$

*For any fixed $M \in \Delta_H$ and every $t \in \{1, \ldots, T\}$, the following two-sided recursion holds:*

$$\Gamma_{t+1}^M \geq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+, \tag{57}$$

$$\Gamma_{t+1}^M \leq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle. \tag{58}$$

*Proof.* From the state updates (13) (for $\pi(M)$) and (39) (for the algorithm), the arrival $w_t$ cancels and we obtain the exact identity

$$\Gamma_{t+1}^M = x_{t+1}^{\pi(M)} - x_{t+1}^{\mathcal{A}} = \alpha\Gamma_t^M - \big(u_t^{\pi(M)} - u_t^{\mathcal{A}}\big). \tag{59}$$

Consequently, bounding $\Gamma_{t+1}^M$ reduces to bounding the control-action difference $u_t^{\pi(M)} - u_t^{\mathcal{A}}$.

Using the SDAC representation (12) for the benchmark control action and the bounds of Lemma 9 for the algorithm control action, we have

$$u_t^{\pi(M)} = \langle M, \widetilde{w}_t\rangle,$$

and

$$\langle M_t, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+ \leq u_t^{\mathcal{A}} \leq \langle M_t, \widetilde{w}_t\rangle.$$

Subtracting the upper bound on $u_t^{\mathcal{A}}$ from $u_t^{\pi(M)}$ yields the lower bound on $u_t^{\pi(M)} - u_t^{\mathcal{A}}$:

$$u_t^{\pi(M)} - u_t^{\mathcal{A}} \geq \langle M, \widetilde{w}_t\rangle - \langle M_t, \widetilde{w}_t\rangle = -\langle M_t - M, \widetilde{w}_t\rangle.$$

Inserting this into the identity (59) gives

$$\Gamma_{t+1}^M \leq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle,$$

which is (58).

Similarly, using the lower bound on $u_t^{\mathcal{A}}$ produces an upper bound on $u_t^{\pi(M)} - u_t^{\mathcal{A}}$:

$$\begin{aligned}u_t^{\pi(M)} - u_t^{\mathcal{A}} &\leq \langle M, \widetilde{w}_t\rangle - \Big(\langle M_t, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+\Big) \\ &= -\langle M_t - M, \widetilde{w}_t\rangle + \alpha[\Gamma_t^{M_t}]_+.\end{aligned}$$

Substituting this bound into the identity (59) yields

$$\Gamma_{t+1}^{M} \geq \alpha \Gamma_t^M + \langle M_t - M, \widetilde{w}_t \rangle - \alpha [\Gamma_t^{M_t}]_+,$$

which is (57) and completes the proof. □

**Lemma 11** (State deviation bound)**.** *For every $t \in \{1, \dots, T\}$, let the state gap be defined by* (56)*. Then*

$$-\frac{\alpha\,\eta L}{(1-\alpha)^3} \leq \Gamma_t^{M_t} = x_t^{\pi(M_t)} - x_t^{\mathcal{A}} \leq \frac{\alpha\,\eta L}{(1-\alpha)^2}.$$

*Proof.* From (58), for any fixed benchmark $M \in \Delta_H$ and every $s \in \{1, \dots, T\}$,

$$\Gamma_{s+1}^{M} \leq \alpha \Gamma_s^M + \langle M_s - M, \widetilde{w}_s \rangle.$$

Applying the recursion repeatedly up to time $t$ yields

$$\begin{aligned}
\Gamma_t^M &\leq \alpha \Gamma_{t-1}^M + \langle M_{t-1} - M, \widetilde{w}_{t-1} \rangle \\
&\leq \alpha^2 \Gamma_{t-2}^M + \alpha \langle M_{t-2} - M, \widetilde{w}_{t-2} \rangle \\
&\qquad + \langle M_{t-1} - M, \widetilde{w}_{t-1} \rangle \\
&\ \ \vdots \\
&\leq \alpha^{t-1} \Gamma_1^M + \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M, \widetilde{w}_s \rangle.
\end{aligned}$$

In this bound, choose the fixed comparison parameter to be $M = M_t$. Since $x_1^{\pi(M_t)} = x_1^{\mathcal{A}} = 0$, we have $\Gamma_1^{M_t} = 0$, and therefore

$$\Gamma_t^{M_t} \leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M_t, \widetilde{w}_s \rangle.$$

Note that $\|\widetilde{w}_s\|_\infty = \max_{i \leq H} \alpha^i w_{s-i} \leq \alpha$ since $w_\tau \in [0,1]$ and $i \geq 1$. Hence,

$$\begin{aligned}
\Gamma_t^{M_t} &\leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \big| \langle M_s - M_t, \widetilde{w}_s \rangle \big| \\
&\leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \|M_s - M_t\|_1 \|\widetilde{w}_s\|_\infty \\
&\leq \alpha\,\eta L \sum_{s=1}^{t-1} \alpha^{t-1-s} (t-s) \\
&= \alpha\,\eta L \sum_{j=1}^{t-1} j\,\alpha^{j-1} \leq \alpha\,\eta L \sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{\alpha\,\eta L}{(1-\alpha)^2}.
\end{aligned}$$

Here we used the geometric-series identity

$$\sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{1}{(1-\alpha)^2}.$$

Similarly, for the lower bound, unrolling (57) with the same fixed comparison parameter $M = M_t$ yields

$$\Gamma_t^{M_t} \geq \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M_t, \widetilde{w}_s \rangle - \sum_{s=1}^{t-1} \alpha^{t-s} [\Gamma_s^{M_s}]_+.$$

The upper bound just proved applies to every earlier time $s$, so $[\Gamma_s^{M_s}]_+ \leq \alpha \eta L/(1-\alpha)^2$. Also, by Hölder's inequality and Lemma 8,

$$\begin{aligned}
\Gamma_t^{M_t} &\geq -\alpha\,\eta L \sum_{s=1}^{t-1} \alpha^{t-1-s} (t-s) \\
&\qquad - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{s=1}^{t-1} \alpha^{t-s} \\
&= -\alpha\,\eta L \sum_{j=1}^{t-1} j\,\alpha^{j-1} - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{j=1}^{t-1} \alpha^j.
\end{aligned}$$

Extending both sums to infinity gives

$$\Gamma_t^{M_t} \geq -\alpha\,\eta L \sum_{j=1}^{\infty} j\,\alpha^{j-1} - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{j=1}^{\infty} \alpha^j.$$

Using the geometric-series identities

$$\sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{1}{(1-\alpha)^2}, \qquad \sum_{j=1}^{\infty} \alpha^j = \frac{\alpha}{1-\alpha},$$

we obtain

$$\Gamma_t^{M_t} \geq -\frac{\alpha\,\eta L}{(1-\alpha)^2} - \frac{\alpha^2\,\eta L}{(1-\alpha)^3} = -\frac{\alpha\,\eta L}{(1-\alpha)^3}.$$

Hence the claim follows. □

**Lemma 12** (Action-state gap)**.** *Let the state gap be defined by* (56)*. Then for all $t \in \{1, \dots, T\}$,*

$$0 \leq u_t^{\pi(M_t)} - u_t^{\mathcal{A}} \leq \frac{\alpha^2\,\eta L}{(1-\alpha)^2}.$$

*Proof.* By (12) and (37), the counterfactual control action equals the nominal control action, $u_t^{\pi(M_t)} = \langle M_t, \widetilde{w}_t \rangle = \widehat{u}_t$, while $u_t^{\mathcal{A}} = \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\} \leq \widehat{u}_t$ by (38). Hence the gap is nonnegative and

$$u_t^{\pi(M_t)} - u_t^{\mathcal{A}} = \widehat{u}_t - \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\} = \big[\langle M_t, \widetilde{w}_t \rangle - \alpha x_t^{\mathcal{A}}\big]_+.$$

As established in the proof of Lemma 9 (using feasibility of $\pi(M_t)$),

$$\big[\langle M_t, \widetilde{w}_t \rangle - \alpha x_t^{\mathcal{A}}\big]_+ \leq \alpha\,[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}]_+ = \alpha\,[\Gamma_t^{M_t}]_+.$$

By Lemma 11, $[\Gamma_t^{M_t}]_+ \leq \dfrac{\alpha\,\eta L}{(1-\alpha)^2}$, so

$$u_t^{\pi(M_t)} - u_t^{\mathcal{A}} \leq \alpha\,[\Gamma_t^{M_t}]_+ \leq \frac{\alpha^2\,\eta L}{(1-\alpha)^2},$$

which completes the proof. □

**Lemma 13** (Stability term)**.** *For all horizons $T$,*

$$\sum_{t=1}^{T} \Big( c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \Big) \leq \frac{\alpha T \eta L^2}{(1-\alpha)^2},$$

*where $L = \alpha L_u + \frac{L_x}{1-\alpha}$.*

*Proof.* By Lipschitz continuity of $c_t$ (see equation (5)),

$$\begin{aligned}
&c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\
&\qquad \leq L_u \big| u_t^{\mathcal{A}} - u_t^{\pi(M_t)} \big| + L_x \big| x_t^{\mathcal{A}} - x_t^{\pi(M_t)} \big|.
\end{aligned}$$

From Lemma 12,

$$|u_t^{\mathcal{A}} - u_t^{\pi(M_t)}| \le \frac{\alpha^2\,\eta L}{(1-\alpha)^2}.$$

Also, by Lemma 11 and (56),

$$|x_t^{\mathcal{A}} - x_t^{\pi(M_t)}| = |\Gamma_t^{M_t}| \le \frac{\alpha\,\eta L}{(1-\alpha)^3}.$$

Therefore, for each $t \in \{1,\dots,T\}$,

$$\begin{aligned} &c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad \le L_u \frac{\alpha^2\,\eta L}{(1-\alpha)^2} + L_x \frac{\alpha\,\eta L}{(1-\alpha)^3}. \end{aligned}$$

Summing over $t = 1,\dots,T$ gives

$$\begin{aligned} &\sum_{t=1}^T \Big( c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \Big) \\ &\quad \le \frac{\alpha\,T\eta L}{(1-\alpha)^2}\left(\alpha L_u + \frac{L_x}{1-\alpha}\right) = \frac{\alpha\,T\eta L^2}{(1-\alpha)^2}, \end{aligned}$$

where we used $L = \alpha L_u + \frac{L_x}{1-\alpha}$ in the final step. □

*Proof of Theorem 1.* Starting from (7), we have the following decomposition of the regret:

$$\begin{aligned} \mathrm{Reg}_{\mathrm{SDAC}}(T) &= \sum_{t=1}^T c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) \\ &\quad - \min_{M\in\Delta_H} \sum_{t=1}^T c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big) \\ &= \sum_{t=1}^T c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^T c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad + \sum_{t=1}^T c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad - \min_{M\in\Delta_H} \sum_{t=1}^T c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big). \end{aligned}$$

We call the first sum the stability term, and the second sum the OMD term. We will bound each term separately.

We bound the OMD term using the OMD update (42) with the unnormalized negentropy $\Psi$ and Bregman divergence $D_\Psi$ from (41). Under the initialization (36), for any benchmark $M^\star \in \Delta_H$ with $W^\star \triangleq \sum_{i=1}^H M^{\star(i)}$, so $W^\star \le 1$,

$$\begin{aligned} &D_\Psi(M^\star \| M_1) \\ &\quad = \sum_{i=1}^H \Big( M^{\star(i)} \log\big((H+1)M^{\star(i)}\big) - M^{\star(i)} + \tfrac{1}{H+1} \Big) \\ &\quad = \sum_{i=1}^H M^{\star(i)} \log M^{\star(i)} + W^\star \log(H+1) - W^\star + \tfrac{H}{H+1}. \end{aligned}$$

Since each $M^{\star(i)} \in [0,1]$, one has $M^{\star(i)} \log M^{\star(i)} \le 0$, and therefore

$$D_\Psi(M^\star \| M_1) \le W^\star \log(H+1) - W^\star + \tfrac{H}{H+1}.$$

The right-hand side is at most $\log(H+1)$ for every $W^\star \in [0,1]$: equivalently,

$$(1-W^\star)\log(H+1) + W^\star - \tfrac{H}{H+1} \ge 0,$$

which holds because the left-hand side is affine in $W^\star$ and nonnegative at the endpoints $W^\star = 0$ and $W^\star = 1$. Hence $D_\Psi(M^\star \| M_1) \le \log(H+1)$. The surrogate costs $\ell_t$ from (40) are convex, and the chosen subgradients satisfy $\|g_t\|_\infty \le L$ by Lemma 3. Hence, the standard OMD regret bound with Bregman divergence $D_\Psi$ on $\Delta_H$ [21] yields

$$\sum_{t=1}^T \ell_t(M_t) - \min_{M\in\Delta_H} \sum_{t=1}^T \ell_t(M) \le \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}.$$

Using (40), we substitute $\ell_t(M) = c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big)$, so that

$$\sum_{t=1}^T \ell_t(M_t) = \sum_{t=1}^T c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big),$$

$$\min_{M\in\Delta_H} \sum_{t=1}^T \ell_t(M) = \min_{M\in\Delta_H} \sum_{t=1}^T c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big).$$

Thus this term becomes

$$\begin{aligned} &\sum_{t=1}^T c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad - \min_{M\in\Delta_H} \sum_{t=1}^T c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big) \\ &\quad \le \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}. \end{aligned}$$

From Lemma 13, the stability term satisfies

$$\begin{aligned} &\sum_{t=1}^T c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &\quad \le \frac{\alpha\,\eta L^2 T}{(1-\alpha)^2}. \end{aligned}$$

Adding the OMD term and the stability term yields the combined bound

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \;\le\; \frac{\log(H+1)}{\eta} \;+\; \frac{\eta L^2 T}{2} \;+\; \frac{\alpha\,\eta L^2 T}{(1-\alpha)^2}.$$

Rearranging by factoring out the $\eta$ term gives

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \;\le\; \frac{\log(H+1)}{\eta} \;+\; \eta L^2 T\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big).$$

For $A, B > 0$, the function $A/\eta + B\eta$ is minimized over $\eta > 0$ at $\eta = \sqrt{A/B}$, where its value is $2\sqrt{AB}$. Taking $A = \log(H+1)$ and $B = L^2 T\big(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\big)$ yields the optimal step size

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2 T\big(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\big)}},$$

and the resulting regret bound

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \;\le\; 2L\sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big) T \log(H+1)}.$$

□

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